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Surname *Van*	**Initials** M *Voorletters* **Stud. No.**

The undersigned hereby confirms that they are familiar with and accepts the examination regulations of the University of Pretoria.
Your cheat sheet must be handed in with this question paper.

Die ondergetekende bevestig hiermee dat hulle bewus is van die eksamenregulasies van die Universiteit van Pretoria en dat hy/sy hom/haar daaraan onderwerp. Jy moet jou formuleblad inhandig met die vraestel.

Signature [redacted] *Handtekening* Date 3 MARCH *Datum*

Class Test 2 **03 / 03 / 2014** ***Klastoets 2***

FSK116

Total/*Totaal*: 23

Question 1/*Vraag 1*

If you are making a very sensitive glass thermometer, which of the following working liquids would you choose? Substantiate your decision.

Indien jy 'n baie sensitiewe glas termometer maak, watter van die volgende vloeistowwe sal jy gebruik? Gee redes vir jou antwoord.

(a) Mercury / *Kwik* – $\alpha = 1.82 \times 10^{-4}\ °C^{-1}$
(b) Alcohol / *Alkohol* – $\alpha = 1.12 \times 10^{-4}\ °C^{-1}$
(c) Petrol / *Petrol* – $\alpha = 9.60 \times 10^{-4}\ °C^{-1}$
(d) Glycerine / *Gliserien* – $\alpha = 4.85 \times 10^{-4}\ °C^{-1}$

[3] 0

I WOULD CHOOSE ALCOHOL – α – $1.12 \times 10^{-4}\ °C^{-1}$. α IS THE LOWEST
$\therefore$ IT EXPANDS THE LEAST ~~WITH CHANGE~~ IN TEMPERATURE

Question 2/*Vraag 2*

A segment of steel railroad track has a length of 30.000 m and a mass of 1800 kg when the temperature is 0.0°C.

'n Stuk staal-spoorlyn het 'n lengte van 30.000 *m en massa van* 1800 *kg wanneer die temperatuur* 0.0 °C is.

$\alpha_{steel} = 11 \times 10^{-6}\ °C^{-1}$ $c_{steel} = 448\ J \cdot kg^{-1} \cdot K^{-1}$

(a) What is the length of the track when the temperature is 40 °C?
Wat is die lengte van die spoorlyn wanneer die temperatuur 40 °C is?

[3] 3

$\Delta L = L_{INITIAL}\,\alpha\,\Delta T = L_i\,\alpha\,(T_f - T_i)$
$= (30\ M)(11 \times 10^{-6}\ °C^{-1})(40\ °C - 0\ °C)$
$= 0{,}0132\ M$ $L_f = L_i + \Delta L$ $\therefore L_f = 30{,}0132\ M$

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(b) What is the percentage change in the volume of the steel track?
Wat is die persentasie verandering in die volume van die staal-spoor?

[3] 0

$\left(\frac{\Delta L}{L_i}\right) \times 100 \quad \left(\frac{0,0132}{30}\right) \times 100 = 0.044\ \%$

(c) How much heat is added to the steel track during this temperature change?
Hoeveel warmte word by die staal-spoor gevoeg tydens hierdie temperatuur-verandering?

[3] 3

$Q = cm\Delta T$
$= (448\ J/kg \cdot K)(1800\ kg)(40\ C - 0\ C)$
$= 32256000\ J = 32256\ kJ$

Question 3/*Vraag 3*
A 1.00 kg block of copper at 20.0 °C is dropped into a large vessel of liquid nitrogen at it's boiling point, 77.3 K. Ignoring any heat exchange to the vessel, determine how much of the liquid nitrogen boils away in the time the copper takes to reach 77.3 K. The specific heat of copper is 387 $J \cdot kg^{-1} \cdot K^{-1}$, the heat of fusion for nitrogen at + 63 K is 2.55×10^4 J/kg, and the heat of vapourization for nitrogen at 77.3 K is 2.01×10^5 J/kg.
'n 1.00 kg blok koper by 20.0 °C word in 'n groot houer vol vloeibare stikstof by die kookpunt van 77.3 K gegooi. Ignoreer enige warmte oordrag na die houer, en bepaal hoeveel stikstof "wegkook" in die tyd wat dit neem vir die koper om 77.3 K te bereik. Die spesifieke warmte van koper is 387 $J \cdot kg^{-1} \cdot K^{-1}$, die smeltingswarmte van stikstof by 63 K is 2.55×10^4 J/kg, en die verdampingswarmte van stikstof by 77.3 K is 2.01×10^5 J/kg.

[5] 3

COPPER: $Q = cm\Delta T$
$= (387\ J/kg \cdot K)(1\ kg)(77.3K - 293,15\ K)$
$= -83533,95\ J$
NITROGEN: $Q = Lm$
$(-83533,95J) = (2,01 \times 10^5\ J/kg)(m)$
$m = -0,41559\ kg$

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Question 4/*Vraag 4*

Water drips from the nozzle of a shower onto the floor 200 cm below. The drops fall at regular (equal) intervals of time. The first drop strikes the floor at the instant that the fourth drop begins to fall. When the first drop strikes the floor, how far below the nozzle is the second drop?

Water drip uit 'n die stortkop en val op die vloer 200 cm na onder. Die druppels val op gereelde (gelyke) tydsintervalle. Die eerste druppel tref die vloer wanneer die vierde druppel begin val. Wanneer die eerste druppel die vloer tref, hoe ver onder die stortkop is die tweede druppel?

[6] 3

TIME IT TAKES FOR DROP 1 TO REACH THE FLOOR:

$x_f = x_i + v_i t + \frac{1}{2} a t^2$ ✓

$(-2\ m) = (0\ m) + (0\ m\cdot s^{-1})(t) + (\frac{1}{2})(-9.8\ m\cdot s^{-2})(t^2)$

$\left(\frac{(-2\ m)}{(\frac{1}{2})(-9.8\ m\cdot s^{-2})}\right) = t^2 = 0.40816$

$t = \pm\sqrt{0.40816}$ $t = 0.6388\ s$ ✓

∴ 1 DROP FALLS EVERY 0.1597 s ∴ $DROP_2$ FALLS

FOR $(0,6388\ s - 0,1597\ s = 0,4791\ s)$

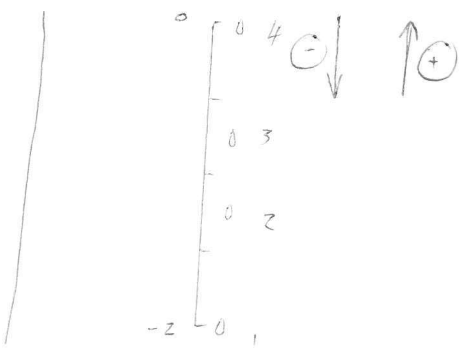
$x_f = x_i + v_i t + \frac{1}{2} a t^2$ ✓

$= (0\ m) + (0\ m\cdot s^{-1})(0,4791\ s) + (\frac{1}{2})(-9,8\ m\cdot s^{-2})(0,4791\ s)^2$

$= -2,34759\ m$

∴ $DROP_2$ IS 2,34759 M BELOW THE NOZZLE

THE END / DIE EINDE



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